



Innovation University

Company Management System

Team Members

Aliaa Mohammed Anes	ID: 24030034
Mai Salah Mohamed	ID: 24030016
Rofida Yaser	ID: 24030141
Nour Nabil Nasif	ID: 24030164
Farha Fathy	ID: 24030058
Basmalla Esmail Radi	ID: 24030343

• Introduction

This project presents a comprehensive database system designed to manage all core operations of a company. The system integrates multiple entities such as employees, departments, projects, clients, payroll, attendance, training programs, and contracts into a unified relational structure. The goal is to improve efficiency, ensure data consistency, and support decision-making.

• System Objectives

Maintain structured employee data- Manage departments and organizational structure-
Track projects and their associated clients- Record attendance and calculate
performance- Manage payroll operations and salaries- Organize training programs-
Ensure data integrity and reduce redundancy

• Business Rules

Company Management System Scenario

A software company wants to build a database system to manage its daily operations.

The system stores information about employees, departments, projects, clients, attendance, payroll, training programs, and contracts.

The company has several departments.

Each department has a unique department ID, department name, and location.

A department can employ many employees, but each employee works in only one department.

For every employee, the system stores personal information such as employee ID, full name(first name, last name), birth date(day,month,year), gender, email, phone number(can have muliple phone numbers).

Each employee belongs to one department.

Employees can work on different projects.

A project has a project ID, project name, budget, start date, and end date.

One employee may work on many projects, and one project may have many employees working on it.

Projects are owned by clients.

**The system stores client information including client ID, client name, email, address, and phone number(can have multiple phone numbers).
One client can own multiple projects, but each project belongs to only one client.**

**The company also records employee attendance.
Each attendance record contains attendance ID, check-in time, check-out time, attendance date, and absence days(derived attribute),late minutes(derived attribute).
Every attendance record must belong to one employee, while one employee can have many attendance records.**

**The attendance information is used by the payroll system to calculate salaries.
The payroll entity stores payroll ID, payment date, deductions, bonuses, and net salary(derived attribute).
Employees with absence days may receive deductions, while employees with extra working hours may receive bonuses.**

**The company provides training programs for employees.
Each training program has a training ID, training name, start date, end date, and cost.
An employee can attend many training programs, and each training program can be attended by many employees.**

**Every employee also has a contract.
The contract stores contract ID, contract type, salary, status, start date, and end date.
Each contract must be associated with one employee, and each employee must have one contract.**

The system helps the company manage employees, projects, salaries, attendance, contracts, trainings, and clients efficiently.

● **Challenges**

- Handling many-to-many relationships using bridge tables.
- Managing derived attributes such as net salary.
- Ensuring referential integrity across multiple tables.
- Designing flexible structure for scalability.